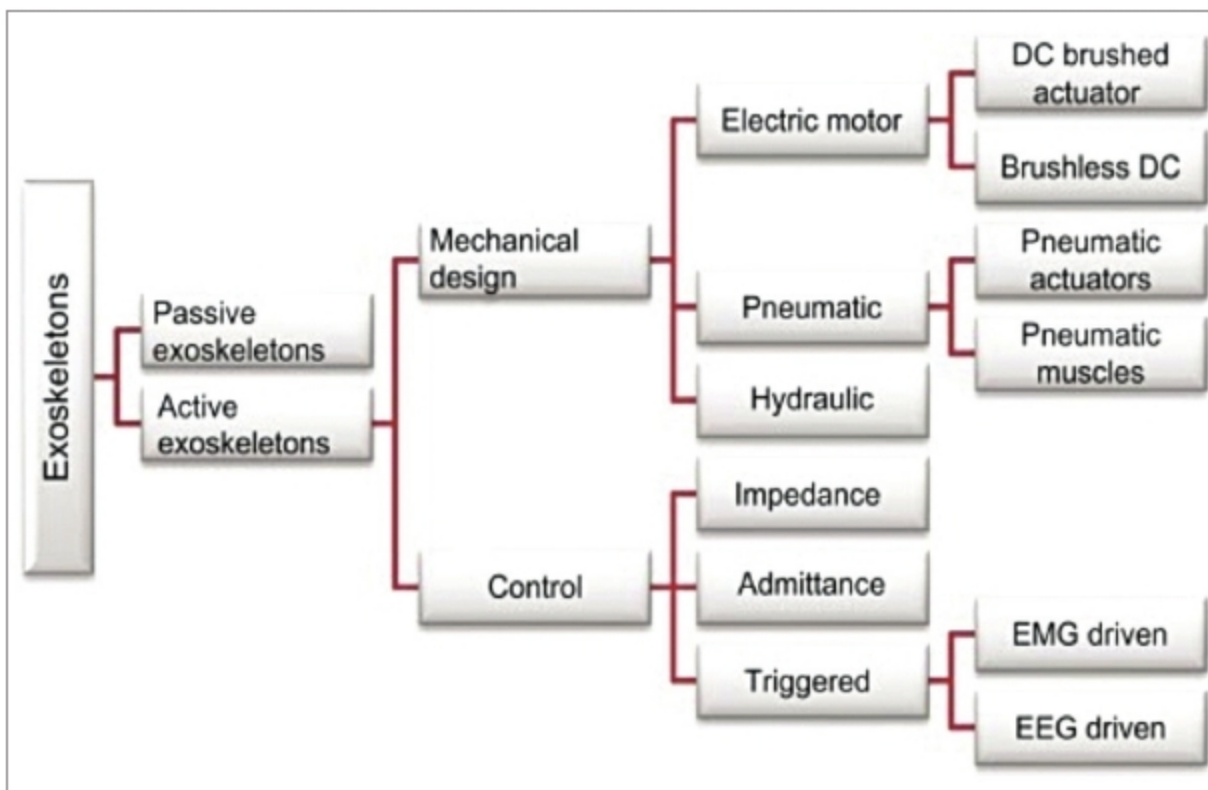




Classification of exoskeletons (Credit: www.sciencedirect.com)

there are mainly two types—passive and active. Passive systems are without actuators, sensors, motors, or batteries, are less expensive and mostly preferred by skilled workers in factories. Since these do not have any electrical power source, motion is controlled with springs and dampers. Take the example of Airframe exoskeleton from the tech company Levitate Technologies. It uses a pulley mechanism instead of electricity.

Powered or active exoskeletons use data from sensors to monitor and respond to persons' movements. Different ways of powering the exoskeletons, including pneumatics, electrical motors, and hydraulics, can be used to improve the strength and endurance of a person's body.

How does it work?

Interaction of exoskeleton with the user involves three main steps—sense, decision, and execution. Fitted with motorised muscles usually, the suit makes it easier to work with heavy objects as it makes them feel much lighter, and hence easier to lift and move.

Advancements in robotics are helping in the creation of intelligent solutions. Predictive algorithms and artificial intelligence allow a faster response from exoskeletons in real time. The cognitive human-robot interaction part consists of algorithms and control strategies to deal with how the user controls the

exoskeleton. Sensing and decision-making are made possible using the information from human operators as well as sensors in the wearables. The focus is on decreasing stress and strain that is usually exerted on muscles and joints.

For physical human-robot interaction, in case of lifting, for example, the weight of arms is transferred to cores of the neck, shoulder, and back for even distribution of energy. When the body part (here, arm) comes back to its resting position, the support is slowly taken off.

These wearables can be made of either rigid materials like carbon fibre or sometimes soft materials such as special kinds of fabric, depending on the application requirements. The actuators made with soft materials are also lightweight, making them useful for such places as the ankle and hand, which require careful movements.

Testing of performance is done via measurement of the interaction forces between the device and the user's limbs during the execution of a task.

Applications

The capabilities and benefits offered by the exoskeletons make them ideal for use in almost every industry, including medical, agriculture, automobile, construction, manufacturing, and so on. These can help people maintain fitness and prevent

health issues as well as help those who face hardships in performing daily tasks.

Their use in industries helps workers avoid chronic health issues. These can also prove to be highly beneficial for first responders and firefighters in disastrous situations. For example, NASA's X1 robotic exoskeleton would allow astronauts to improve their exercises by the addition of resistance force. The data can be recorded and sent to Earth, where doctors can provide feedback.

As per a World Health Organization report, one billion people or fifteen per cent of the total world population experiences some form of disability. These people are more vulnerable to adverse socio-economic experiences. In healthcare, the exosuits are mainly needed in rehabilitation and therapeutic applications to improve the quality of life of patients.

Exosuits are designed for multiple medical conditions, including stroke, cerebral palsy (CP), incomplete spinal cord injury, multiple sclerosis (MS), traumatic brain injury (TBI), and many more. These help patients in regaining limb strength and perform day-to-day tasks that involve walking or using their arms. Exosuits enable paraplegics to stand up and walk at a consistent pace again. These can help in dealing with obesity and balancing body parameters caused by increased sitting time and decreased level of physical activity.

On-board sensors enable accurate tracking of a patient's progress over time; the data is valuable in ensuring a balanced approach. This reduces the work of therapists as assistance can be simply customised to the needs of individual patients.

During body movements, nerve signals are sent from the brain to muscles via motor neurons. To give it a more natural feel, bio-sensors are attached nowadays so that a wearer's intention can be understood in advance from the bio-signals received, and response from exoskeleton can be simultaneous. Researchers are also working on electroencephalogram (EEG) bio-sensors that can measure activity across the